

Demand Investigation Flow

A simple guide to understanding the Investigation Layer

Main idea: Investigation is not just calculating numbers. It is a controlled process that moves from **What changed?** → **Where did it change?** → **What explains it?** → **What evidence supports the explanation?**.

1. The Complete Flow

Step	Main question	Purpose
1. Question Understanding	What is the user asking?	Identify KPI, period and comparison.
2. Change Detection	What changed?	Measure current vs previous period.
3. Dimension Analysis	Where could the change be?	Compare dimensions such as Region, Customer, Product.
4. Contribution Analysis	Who contributed most?	Rank segments by contribution to the movement.
5. Hierarchical Drill-down	Where exactly should I look deeper?	Move through the hierarchy into the strongest branches.
6. Driver Investigation	Why might it have changed?	Test business factors such as price, promotion, seasonality, stock and comp
7. Evidence Layer	What proves/supports this?	Attach measurable evidence to each finding.
8. Explanation	How do we tell the story?	Turn analytical results into a transparent business explanation.

2. Change Detection — What happened?

The system first establishes that a meaningful change exists. For example:

June Demand = 120,000
July Demand = 150,000
Change = +30,000 (+25%)

At this point, the system knows **what happened**, but not where or why.

3. Dimension Analysis — Where did it happen?

The system examines dimensions that can split the KPI: Region, Customer, Product, Channel, etc. It looks for dimensions where the segments behave differently enough to explain the overall movement.

Important distinction: Dimension Analysis selects useful ways to partition the change; it does not yet explain the business cause.

4. Contribution Analysis — Who contributed most?

Once a dimension is selected, the system measures how much each segment contributed to the total movement. Example: Cairo +15k, Canal +8k, Delta +4k. This tells us **which segments deserve attention first**.

Contribution Analysis = ranking the sources of movement.

5. Hierarchical Drill-down — Where should we investigate deeper?

Drill-down turns the investigation into a tree. A parent node is broken into children, and only meaningful branches are expanded. Example hierarchy:

Company → Region → Customer → Product → Date

If Cairo caused a large part of the change, the system can investigate Cairo's customers; then the strongest customer can be investigated by product.

Expand means: spend more computation to investigate inside this branch. **Stop** means: keep the result but do not go deeper.

6. Branching: why 3, 2, 2?

If the configuration is **BRANCHING_BY_DEPTH = (3, 2, 2)**, it means the allowed number of branches to expand changes by depth: 3 at the first relevant level, then 2, then 2. It does **not** mean 'drop three nodes'.

There are two separate decisions:

- **drillable()**: Is this branch eligible for deeper investigation?
- **frontier_for()**: Among eligible branches, which ones do we actually expand?

A branch can stop because of contribution threshold, maximum depth, no remaining dimensions, an 'Other' bucket, or because it was not selected within the branching budget.

7. Driver Investigation — Why did it change?

After drill-down identifies the important entity, the system changes the question from **WHERE?** to **WHY?**. Candidate drivers can include:

Driver	What to check
Price	Did price increase/decrease?
Promotion	Was a promotion active?
Seasonality	Is the period normally high/low?
Holiday/Event	Did Ramadan, Eid, Black Friday or another event affect demand?
Competitor	Did competitor price/availability change?
Availability	Was demand constrained by stock?
Customer behavior	Did a specific customer/order pattern change?

Important: a driver being correlated with a change does not automatically prove causality. The system should describe the result as an observation, association, or model-attributed driver unless stronger causal evidence exists.

8. Evidence Layer — How confident are we?

Every important claim should be connected to measurable evidence. For example:

Claim: P004 demand increased because promotion was active.

Evidence: Promotion OFF → ON; price 100 → 85; historical promoted periods show higher demand; current demand increased +31%.

The Evidence Layer should make the investigation **traceable**: a reviewer can move from the conclusion back to the numbers and analytical steps that produced it.

9. Validation — Does the investigation make sense?

The tree should be checked for consistency. If a parent movement is +10,000 and the selected children explain only +8,000, the system should report that 20% remains unexplained rather than pretending the explanation is complete.

Tree validation also needs to distinguish legitimate cases such as a pure split or a truncated top-N view from genuine calculation drift.

10. Full Example

User question: Why did demand increase in July compared with June?

Step	Stage	What the system learns
1	Change Detection	June = 120k, July = 150k → demand increased by +30k (+25%).
2	Dimension Analysis	Region has strong explanatory power, so investigate Region.
3	Contribution Analysis	Cairo +15k, Canal +8k, Delta +4k. Cairo is the largest contributor.
4	Drill-down	Inside Cairo, investigate customers. C001 +8k and C005 +3k are the strongest branches.
5	Deeper Drill-down	Inside C001, P004 +5k is the strongest product contributor.
6	Driver Investigation	Check P004 price, promotion, seasonality, holidays, competitor price and stock.
7	Evidence	Attach the observed price/promotion changes and historical comparisons to the driver claim.
8	Conclusion	Explain what changed, where it happened, which drivers have evidence, and what remains uncertain.

The investigation path

Company +30k (+25%)
→ Cairo +15k (largest regional contributor)
→ C001 +8k (largest Cairo contributor)
→ P004 +5k (largest C001 contributor)
→ investigate Price / Promotion / Seasonality / Holiday / Competitor / Stock
→ attach evidence
→ produce final explanation

11. The Mental Model to Remember

Change Detection = WHAT happened?

Dimension Analysis = WHERE can I split the change?

Contribution Analysis = WHO contributed most?

Hierarchical Drill-down = WHERE exactly should I go deeper?

Driver Investigation = WHY might it have changed?

Evidence Layer = WHAT supports the claim?

Validation = DOES the investigation reconcile and remain honest?

The most important architectural principle: the analytical engine should calculate facts and make structured decisions; the future AI Analyst should orchestrate those capabilities and explain their results, rather than inventing the analysis itself.